KYRIAKOS STYLIANOPOULOS

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Kyriakos Stylianopoulos

PhD Candidate

Research Interests – Machine Learning, Bayesian Statistics, Reinforcement Learning, Wireless Communications, Reconfigurable Intelligent Surfaces, Observational Astronomy

EDUCATION

PhD – Informatics & Telecommunications

National & Kapodistrian University of Athens

December 2021 – currently | Athens, Greece

- Dissertation title: *"Dynamic Control and Over-the-Air Computing with Reconfigurable Metasurfaces"*
- Advised by Prof. G. Alexandropoulos, NoesysLab.
- Participation in three Horizon 2020 projects and one European Space Agency (ESA) tender, co-authoring 30 deliverables.

MPhil – Physics

University of Cambridge, Selwyn College

October 2019 – October 2020 | Cambridge, United Kingdom

- Master by research, affiliated with the Centre for Scientific Computing, Cavendish Laboratory and the Institute for Astronomy.
- Thesis Title: *"Machine Learning Applied to Gaia and Other Survey Data: Applications Supporting a Polarisation Survey"*.
- Supervisors: Prof. Gerry Gilmore FRS (principal), Prof. Kaisey Mandel, and Dr. Philip Blakely.
- Developed machine learning methods for processing large survey data. Produced catalogues of candidate galaxies, quasars / active galactic nuclei, chemically peculiar stars, and sources of variable brightness from uncategorised observations.

BSc – Informatics & Telecommunications

National & Kapodistrian University of Athens

October 2014 – June 2019 | Athens, Greece

- GPA: *8.39/10*.
- Specialisations: Data and Knowledge Management, Software
- Thesis Title: *"A Novel Probability-based Data Clustering Application for Detecting Elongated Clusters with Application to the Line Detection Problem"*.
- Supervisors: Prof. S. Theodoridis, and Dr. K. Koutroumbas, National Observatory of Athens.
- Published a novel clustering algorithm for edge and line detection problems in image processing.

RESEARCH OUTPUT

- *25* publications – first or joint-first author in 16.
- *446* total citations – *H-index: 10*.
- Top venues: *Proc. IEEE* (25.9 IF), *TWC* (10.7 IF), *ICASSP 2022, 2023 & 2026* (A1 Rank), *ICC 2022 & 2025* (B Rank).
- *177* research interest score (*ResearchGate.com* – 25 million members) – top *5%* compared by year of first publication.
- Inputs to European Telecommunications Standards Institute (ETSI):
  1. ISC(24)000118: "Data Representation for Sensing"
  2. ISC(25)000072: "Processing of sensing signals"

ACADEMIC EXPERIENCE

Visiting Graduate Researcher

University of Surrey, United Kingdom

October 2023 – November 2023

- Awarded the "Greek-British Short Term Mobility Scholarship 2023" for research visit.

Research Assistant

UoA - NoesysLab, Greece

February 2021 – December 2021

- Worked under the RISE-6G H2020 project.

Exchange Student (ERASMUS/SEMP)

Università Della Svizzera Italiana, Switzerland

September 2016 – February 2017

- Taken four undergraduate and graduate courses with a semestral GPA of *9.08/10*.

VARIOUS EXPERIENCE

- Trainee of ESA Academy on Satellite Communications Systems course (March 2026).
- Mentor at NASA Space Apps challenge 2022.
- Invitee on Google FooBar Challenge – participated in and completed 3 out of 5 stages of algorithmic problems (February – May 2021).
- Certified Tensorflow developer (DeepLearning.AI & Google through Coursera - 4 courses, 2021).
- Backend maintainer on USI's ASQ platform as extra credits (July – August 2017).
- Teaching assistant in three programming courses (February 2016 – June 2017).
- Involvement in a location-based social network startup (February – October 2016).
- Contributor to open source projects (AstroML, ASQ) and communities (StackOverflow).

TECHNOLOGIES

C | Python | C++ | Java | MATLAB | PyTorch/Tensorflow | MPI | CUDA | Scikit-Learn | LaTeX | Web technologies

SPOKEN LANGUAGES

Greek	Native	●●●●●●●
English	C2/IELTS	●●●●●●○
French	B2	●●●●○○○

LIST OF PUBLICATIONS

Journals

[J9] G. Stamatelis, **K. Stylianopoulos**, and G. C. Alexandropoulos, "Learning to Distributedly Estimate under Partially Known Dynamics: A Covariance-Agnostic Neural Kalman Consensus Filter", in *IEEE Trans. Signal Process.*, 2026, under review.

[J8] **K. Stylianopoulos**, M. E. Pandolfo, P. Di Lorenzo, and G. C. Alexandropoulos, "Metasurfaces-integrated wireless neural networks for lightweight over-the-air edge inference" *IEEE Wireless Commun.*, 2026, Early Access.

[J7] **K. Stylianopoulos** and G. C. Alexandropoulos, "Universal approximation with XL MIMO systems: OTA classification via trainable analog combining," *IEEE Wireless Commun. Lttr.*, 2026, under review.

[J6] **K. Stylianopoulos**, P. Di Lorenzo, and G. C. Alexandropoulos, "Over-the-air edge inference via metasurfaces-integrated artificial neural networks," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 13818-13834, 2026.

[J5] G. Stamatelis, **K. Stylianopoulos**, and G. C. Alexandropoulos, "Evolving multi-branch attention convolutional neural networks for online RIS configuration," *IEEE Trans. Cogn. Commun. Netw.*, vol. 12, pp. 14-28, 2026.

[J4] F. Saggese, V. Croisfelt, **K. Stylianopoulos**, G. C. Alexandropoulos, and P. Popovski, "Control plane for reconfigurable intelligent surfaces," *IEEE Commun. Standards Mag.*, vol. 8, no. 4, pp. 24–30, Dec. 2024.

[J3] F. Saggese, V. Croisfelt, R. Kotaba, **K. Stylianopoulos**, G. C. Alexandropoulos, and P. Popovski, "On the impact of control signaling in RIS-empowered wireless communications," *IEEE Open J. Commun. Society*, vol. 5, pp. 4383–4399, 2024.

[J2] G. C. Alexandropoulos, **K. Stylianopoulos**, C. Huang, C. Yuen, M. Bennis, and M. Debbah, "Pervasive machine learning for smart radio environments enabled by reconfigurable intelligent surfaces," *Proc. IEEE*, vol. 110, no. 9, pp. 1494–1525, Sep. 2022.

[J1] **K. Stylianopoulos** and K. Koutroumbas, "A probabilistic clustering approach for detecting linear structures in two-dimensional spaces," *Pattern Recognit. Image Anal.*, vol. 31, pp. 671–687, 2021.

Conferences

[C15] **K. Stylianopoulos**, M. Fabiani, G. Torcolacci, D. Dardari, and G. C. Alexandropoulos, "Over-the-air extreme learning machines with stacked intelligent metasurfaces," in *Proc. IEEE Int. Workshop Signal Process. Adv. Wireless Commun. (SPAWC)*, Athens, Greece, Sept. 2026, to be presented.

[C14] F. Costanzo, S. Mekki, P. Mursia, **K. Stylianopoulos**, G. Zafzouf, A. Al Khansa, T.-T. Luc, B. Denis, G. Madhusudan, M. Crozzoli, G. C. Alexandropoulos, and H. Wymeersch, "Distributed intelligent sensing and communications for 6G: Architecture and use cases," in *Proc. Joint Eur. Conf. Netw. Commun. & 6G Summit (EuCNC/6G Summit)*, Poznan, Poland, 2025, pp. 583-588.

[C13] **K. Stylianopoulos**, M. Fabiani, G. Torcolacci, D. Dardari, and G. C. Alexandropoulos, "Over-the-air extreme learning machines with XL reception via nonlinear cascaded metasurfaces," in *Proc. Int. Zurich Seminar Inf. Commun. (IZS)*, Zurich, Switzerland, Feb. 2026, pp. 148–153.

[C12] M. E. Pandolfo, **K. Stylianopoulos**, G. C. Alexandropoulos, and P. Di Lorenzo, "Over-the-air semantic alignment with stacked intelligent metasurfaces," in *Proc. Eur. Signal Process. Conf. (EUSIPCO)* , 2026, to be presented.

[C11] **K. Stylianopoulos** and G. C. Alexandropoulos, "Integrating stacked intelligent metasurfaces and power control for dynamic edge inference via over-the-air neural networks," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, Barcelona, Spain, May 2026.

[C10] P. Gavriilidis, **K. Stylianopoulos**, and G. C. Alexandropoulos, "MIMO communications with 1-bit RIS: Asymptotic analysis and over-the-air channel diagonalization," in *Proc. IEEE Asilomar Signals, Syst. Comput. Conf.*, Pacific Grove, USA, Oct. 2025, pp. 1–5.

[C9] G. Stamatelis, **K. Stylianopoulos**, and G. C. Alexandropoulos, "Learning RIS configuration with quantized responses: A neuroevolution-trained multi-branch attention convolutional neural network," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, Montreal, Canada, 2025, pp. 1894–1899.

[C8] **K. Stylianopoulos**, G. Madhusudan, G. Jornod, S. Mekki, F. Costanzo, H. Chen, P. Mursia, M. Crozzoli, E. C. Strinati, G. C. Alexandropoulos, and H. Wymeersch, "Distributed intelligent sensing and communications for 6G: Architecture and use cases," in *Proc. Joint Eur. Conf. Netw. Commun. & 6G Summit (EuCNC/6G Summit)*, Poznan, Poland, 2025, pp. 583–588.

[C7] **K. Stylianopoulos**, P. Gavriilidis, G. Gradoni, and G. C. Alexandropoulos, "Graph-CNNs for RF imaging: Learning the electric field integral equations," in *Proc. Eur. Signal Process. Conf. (EUSIPCO)*, Palermo, Italy, Sep. 2025.

[C6] X. Ma, Y. Zhou, Q. Luo, Y. Ma, **K. Stylianopoulos**, and G. C. Alexandropoulos, "1-bit subTHz RIS with planar tightly coupled dipoles: Beam shaping and prototypes," in *Proc. Eur. Conf. Antennas Propag. (EuCAP)*, Glasgow, United Kingdom, 2024, pp. 1–5.

[C5] **K. Stylianopoulos**, M. Bayraktar, N. González-Prelcic, and G. C. Alexandropoulos, "Autoregressive attention neural networks for non-line-of-sight user tracking with dynamic metasurface antennas," in *Proc. IEEE Int. Workshop Comput. Adv. Multi-Sensor Adaptive Process. (CAMSAP)*, 2023, pp. 391–395.

[C4] **K. Stylianopoulos**, M. Merluzzi, P. Di Lorenzo, and G. C. Alexandropoulos, "Lyapunov-driven deep reinforcement learning for edge inference empowered by reconfigurable intelligent surfaces," *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, Rhodes Island, Greece, 2023, pp. 1–5.

[C3] **K. Stylianopoulos** and G. C. Alexandropoulos, "Online RIS configuration learning for arbitrary large numbers of 1-bit phase resolution elements," in *Proc. IEEE Int. Workshop Signal Process. Adv. Wireless Commun. (SPAWC)*, Oulu, Finland, 04-06 Jul 2022.

[C2] **K. Stylianopoulos**, G. Alexandropoulos, C. Huang, C. Yuen, M. Bennis, and M. Debbah, "Deep contextual bandits for orchestrating multi-user MISO systems with multiple RISs," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Seoul, South Korea, 2022, pp. 1556–1561.

[C1] **K. Stylianopoulos**, N. Shlezinger, P. del Hougne, and G. C. Alexandropoulos, "Deep-learning-assisted configuration of reconfigurable intelligent surfaces in dynamic rich-scattering environments," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, Singapore, Singapore, 2022, pp. 8822–8826.

Book Chapters

[B1] **K. Stylianopoulos**, P. Di Lorenzo, and G. C. Alexandropoulos, "Over-the-Air Goal Oriented Communications," in the book *"Smart Wireless Environments for Integrating Sensing and Communications,"* Springer Nature, to appear, 2026; edited by K. Meng, C. Masouros, and G. C. Alexandropoulos.